

Karan Singh Sodhi – Software Engineer

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Summary

Full-stack Software Engineer focused on building **scalable real-time applications** using MERN and Next.js. Developed a Discord-like platform with **real-time chat, voice/video (LiveKit), and role-based systems**. Strong in **data structures, RESTful API design, object-oriented programming, and backend performance optimization**. Experienced with WebSocket-based systems using Socket.IO. Comfortable working in agile environments, debugging complex issues, and delivering production-ready software.

Experience

Web Developer Intern, V'NSPIRE – Agra, India

- Engineered and deployed a **production-grade MERN application** with authentication, posting, and feed workflows, supporting dynamic user-generated content.
- Implemented **real-time messaging** using Socket.IO, enabling low-latency communication for concurrent users.
- Designed and optimized **MongoDB data models** for users, posts, and messages, scaling to **1,000+ records** with efficient query performance.
- Developed and exposed **10+ RESTful APIs** using Node.js and Express with structured validation, error handling, and clear service contracts.
- Integrated **JWT-based authentication and role-based authorization**, securing protected endpoints and ensuring controlled access.
- Optimized frontend rendering and state management to **improve UI responsiveness and reduce unnecessary re-renders**.

Projects

Real-Time Collaboration Platform (Discord Clone) *Next.js 15, Socket.IO, LiveKit, PostgreSQL* [Live](#) | [Code](#)

- Engineered a **scalable real-time collaboration platform** with chat, voice, and video using WebSockets and SFU-based architecture (LiveKit), enabling low-latency communication.
- Designed and optimized **PostgreSQL schemas** with indexing, constraints, and efficient query patterns to improve data retrieval performance by **25%**.
- Implemented **secure authentication and role-based access control (RBAC)** for multi-server environments, ensuring isolated and secure user interactions.
- Optimized API performance and event handling and state synchronization to **reduce client-side re-renders by 30%** and improve overall responsiveness.

Scalable E-Commerce Platform (E-Kart) *MongoDB, Express.js, React.js, Node.js, Stripe, Cloudinary* [Live](#) | [Code](#)

- Architected and deployed a **scalable full-stack eCommerce platform** using MERN, serving dynamic content with modular REST APIs and maintainable service-based architecture.
- Designed and implemented **JWT-based authentication and role-based authorization (RBAC)**, securing protected routes and reducing unauthorized access risks.
- Integrated **Stripe payment gateway** with end-to-end order validation, ensuring reliable transactions and consistent checkout flows.
- Optimized MongoDB indexing and query execution to **reduce API response latency by 30%** and improve overall application performance.

Education

Dr. A.P.J. Abdul Kalam Technical University

08/2023 – 08/2027

Bachelor of Technology in Computer Science and Engineering

Skills

Languages: C++, JavaScript, Python

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, WebSockets (Socket.IO), JWT Authentication, System Design

Databases: PostgreSQL, MongoDB, Indexing, Query Optimization

Core CS & Tools: Data Structures, Algorithms, OOP, Operating Systems, Git, GitHub, Docker, Postman, Stripe, Cloudinary

Certifications

- Full Stack Web Development (MERN)** — Sheriyans Coding School
- JavaScript (ES6+) & Data Structures** — Coursera
- AWS Cloud Practitioner & Fundamentals** — freeCodeCamp